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PERFORMANCE SPECIFICATION

VALVE, SOLENOID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of 24-volt (V), solenoid actuated, hydraulic directional control valves with an operating pressure of 3000 pounds per square inch (psi) (20 685 kilopascals (kPa)) (see 6.1).

1.2 Classification. Valves should be of the following types, as specified (see 6.2):

Type I	- Two position, four-way, rated flow capacity: 6 gallons per minute (gal/min) (0.38 liters per second (L/s)).
Type II	- Two position, two-way, rated flow capacity: 3.5 gal/min (0.22 L/s).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

AMSC N/A FSC 2590

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-462	- Measurement of Electromagnetic Interference Characteristics, Test Methods Standard for (see 4.3.1).
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines (see 4.3.1).

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified. the issues are those cited in the solicitation.

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR)- Title 10, Parts 30 and 40.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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DRAWINGS

DEPARTMENT OF DEFENSE

9008233	- Valve, Solenoid (Type I) (Interface).
9098673	- Valve, Solenoid (Type II) (Interface).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117	- Standard Practice for Operating Salt Spray (Fog) Apparatus (DoD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), first article samples shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Unless otherwise specified, the design, materials, and manufacturing process selection shall be the prerogative of the contractor as long as all articles submitted to the Government fully meet the operating, interface, ownership and support, and operating environment requirements specified. (Asbestos, cadmium and radioactive materials shall not be used in this item. Radioactive material is defined by Title 10, CFR, Parts 30 and 40, and other radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries).

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3.2.1 Metals. All metals used in valve construction, except metal which is in constant contact with the hydraulic fluid, shall be of corrosion-resisting type or shall be adequately protected to resist corrosion during the normal service life of the valve and shall meet the requirements of salt spray 3.6.4.

3.2.1.1 Dissimilar metals. Dissimilar metals shall be protected from galvanic corrosion.

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. The valve shall meet the following performance requirements.

3.3.1 Operating time.

3.3.1.1 Type I. The valve shall operate (change port interconnection) in not more than 0.05 second (s), with 18 V direct current (dc) applied.

3.3.1.2 Type II. The valve shall operate (change port interconnection) in not more than 0.07 s, with 18 Vdc applied.

3.3.2 Operating pressure. The valve shall operate with any pressure from 250 psi (1724 kPa) applied to any port except return, and shall operate at any pressure within a pressure differential range, between the pressure port and return port, from 250 psi to 3000 psi (1724 kPa to 20 685 kPa) (return port applicable to type I only) while using a conforming hydraulic fluid (see 6.3).

3.3.3 Operating voltage. The valve shall operate at any voltage within a range of 18 to 30 Vdc, with operating pressure applied.

3.3.4 Operating current. The valve shall require no more than 1.0 ampere (A) at 24 Vdc to operate.

3.3.5 Rated flow and pressure drop.

3.3.5.1 Type I. The rated flow capacity of the valve using hydraulic fluid (see 6.3) shall be 6 gal/min (0.38 L/s), through any interconnected ports, with a pressure drop across the valve to be not more than 150 psi (1034 kPa).

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3.3.5.2 Type II. The rated flow capacity of the valve using hydraulic fluid (see 6.3) shall be 3.5 gal/min (0.22 L/s) with a pressure drop across the valve to be not more than 60 psi (413.7 kPa).

3.3.6 Proof pressure.

3.3.6.1 Type I. The valve shall withstand a proof pressure of 4500 psi (31 026 kPa), except the return port shall withstand a proof pressure of 1500 psi (10 342 kPa), without leakage or damage.

3.3.6.2 Type II. The valves shall withstand a proof pressure of 4500 psi (31 026 kPa) without external leakage or damage.

3.3.7 Burst pressure.

3.3.7.1 Type I. The valve ports shall withstand hydraulic pressure of 7500 psi (51 710 kPa), except the return port shall withstand pressure of 2500 psi (17 237 kPa) without bursting.

3.3.7.2 Type II. The valve shall withstand hydraulic pressure of 7500 psi (51 710 kPa) without bursting.

3.3.8 Leakage.

3.3.8.1 Pressure port.

3.3.8.1.1 Type I. The valve, in the de-energized position, shall withstand hydraulic pressures up to and including 3000 psi (20 685 kPa) applied at the pressure port. Resultant combined internal leakage from cylinder number 2 port and return port shall be not greater than 0.0035 cubic feet per minute (ft^3/min) (10 cubic centimeters per minute (cm^3/min)), with no external leakage.

3.3.8.1.2 Type II. The valve, in the energized position, shall withstand hydraulic-pressures up to and including 3000 psi (20 685 kPa), applied at one port, with resultant internal leakage from the other port not in excess of 0.0018 ft^3/min (5 cm^3/min).

3.3.8.2 Return port (type I only). The valve, in the de-energized position, shall withstand application of hydraulic pressure of 10 psi (69 kPa) at the return port and cylinder number 2 port without external leakage.

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3.3.9 Solenoid temperature rise. The valve solenoid shall not exceed a temperature rise greater than 39 degrees Fahrenheit (°F) (3.9 degrees Celsius (°C)) per minute of continuous operation, when used with voltages specified herein.

3.3.10 Dielectric strength. The solenoid shall withstand application of 1000 V root mean square (rms), (60 hertz (Hz)) potential, between terminals and ground, without flashover. Insulation shall evidence no loosening, cracking, charring, burning, smoking, rupture, or other damage resulting in a reduction in dielectric strength.

3.4 Interface requirements.

3.4.1 Overall envelope and interfaces. The valve shall fit into the hydraulic system with the loader-transporter, guided missile (Hawk Loader) in accordance with the envelope dimensions and the electrical and mechanical interfaces locations and dimensions of drawings 9008233 or 9098673.

3.4.2 Weight. The gross weight of the valve shall not exceed 2 pounds (lb) (0.9 kilogram (kg)).

3.4.3 Solenoid. The solenoid for operating the hydraulic valve shall be compact, designed for continuous or intermittent duty, and shall be provided with single coil windings. The solenoid shall be totally enclosed and shall be so designed that hydraulic fluid can at no time come in contact with the electrical windings.

3.4.4 Coils. The coils shall be evenly wound and insulated to meet the performance requirements specified herein. The coils shall be completely insulated from all other parts of the valve. Coils shall be suitably taped and impregnated, as required, to prevent damage under prolonged exposure to environmental conditions.

3.4.5 Radio interference suppression. The valve, solenoid, shall conform to electromagnetic compatibility requirements of figures 1 and 2.

3.5 Ownership and support requirements.

3.5.1 Endurance. The valve shall withstand 20000 cycles of operation at 3000 psi (20 685 kPa) hydraulic pressure, without damage. A cycle of operation shall consist of operation of the valve from the de-energized position to the energized position, and back to the de-energized position. The valve shall subsequently meet the requirements of 3.3.2 through 3.3.6, and 3.3.8.

3.5.2 Safety. Valves shall be physically and electrically safe to operate.

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3.5.3 Interchangeability. Valves shall be repairable, or functionally and physically interchangeable with respect to like assemblies, sub-assemblies and replaceable parts of the same type and design, regardless of manufacturer.

3.5.4 Identification. Prominently display the valve designator on the outer surface of the valve, and shall include as a minimum the following:

a. Nomenclature “Valve, Solenoid”.
b. Military part number.
c. When required, serial numbers shall be defined (see 6.2).
d. Manufacturer’s CAGE code and name.
e. Date of manufacture.
f. Contract or order number.
g. Symbol “US”.

3.6 Operating environment requirements. The valve shall operate under the following environmental conditions, without degradation in performance.

3.6.1 High temperature. The valve shall withstand exposure to a high temperature of 160 ± 5°F (71 ± 3°C), without damage, and shall conform to performance requirements of 3.3.2, 3.3.3, 3.3.5, and 3.3.8 during exposure. Subsequent performance shall conform to requirements of 3.5.1.

3.6.2 Low temperature. The valve shall withstand exposure to a low temperature of -65 (+5, -0°F) (-54, (+3, -0°C)), without damage, and shall conform to requirements of 3.3.2. 3.3.3, 3.3.5, and 3.3.8 during exposure. Subsequent performance shall conform to the requirements of 3.5.1.

3.6.3 Fungus. The valve shall withstand exposure to fungus. Subsequent performance shall conform to 3.3.2, 3.3.3, 3.3.5, and 3.3.8.

3.6.4 Salt spray. The valve shall withstand exposure to salt spray, with subsequent performance conforming to 3.3.3, 3.3.5, and 3.3.8.

3.6.5 Sand and dust. The valve shall withstand exposure to sand and dust, without damage. Subsequent performance shall conform to 3.3.2, 3.3.3, 3.3.5, and 3.5.1.

3.6.6 Shock. The valve shall withstand exposure to shock, without damage or loosening of components. Subsequent conformance shall conform to 3.3.2, 3.3.3, 3.3.5, and 3.3.6.

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3.6.7 Vibration. The valve shall withstand vibration, without loosening of components. Vibration shall be applied in the direction of each of the three major axes, without damage or loosening of components. Subsequent performance shall conform to 3.3.2, 3.3.3, 3.3.5, and 3.5.1.

3.6.8 Humidity. The valve shall withstand exposure to humidity in and with subsequent performance conformance in 3.3.5 and 3.3.4.

4. VERIFICATION

4.1 Classification of inspection:

a. First article inspection.
    1. Preproduction inspection.
    2. Initial production inspection.
b. Conformance inspections.

4.1.1 First article inspection. When first article inspection is required (see 6.2 and 6.5), it shall be performed on preproduction or initial production samples as specified herein.

4.1.1.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of four valves. The preproduction inspection shall consist of the examinations and tests of table I.

4.1.1.2 Initial production inspection. When specified (see 6.2), the initial production inspection shall consist of selecting four valves, from the first ten valves produced. The valves shall be subjected to the examinations and tests of table I.

4.1.2 Conformance inspection. Conformance inspection shall include those examinations and tests from table I as defined in the contract.

4.2 Order of inspection. The inspection sequence shall be in the order specified in table I.

4.3 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previously qualified designs.

4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipments including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by this specification.

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TABLE I. Verification methods.

Title	Requirements	Verification	Sample No.
Materials & dissimilar metals, overall envelope & interfaces, weight, solenoid, coils, safety, interchangeability & identification	3.2, 3.4.1 thru 3.4.9, 3.5.2 thru 3.5.4	4.3, 4.3.4.1 thru 4.3.4.4, 4.3.5.2, 4.3.5.3, & 4.3.5.4	1 thru 4
Operating time	3.3.1	4.3.3.1	1
Operating pressure	3.3.2	4.3.3.2	
Operating voltage	3.3.3	4.3.3.3	
Operating current	3.3.4	4.3.3.3	
Rated flow and pressure drop	3.3.5	4.3.3.4	
Solenoid temperature rise	3.3.9	4.3.3.8	
Electromagnetic compatibility	3.4.5	4.3.4.5	
Dielectric strength	3.3.10	4.3.3.9	
Burst pressure	3.3.7	4.3.3.6	
Endurance	3.5.1	4.3.5.1	2
Operating pressure	3.3.2	4.3.3.2	
Operating voltage	3.3.3	4.3.3.3	
Operating current	3.3.4	4.3.3.3	
Rated flow and pressure drop	3.3.5	4.3.3.4	
Proof pressure	3.3.6	4.3.3.5	
Leakage	3.3.8	4.3.3.7	
High temperature	3.6.1	4.3.6.1	3
Operating pressure	3.3.2	4.3.3.2	
Rated flow and pressure drop	3.3.5	4.3.3.4	
Proof pressure	3.3.6	4.3.3.5	
Low temperature	3.6.2	4.3.6.2	
Operating pressure	3.3.2	4.3.3.2	
Rated flow and pressure drop	3.3.5	4.3.3.4	
Proof pressure	3.3.6	4.3.3.5	
Sand and dust	3.6.5	4.3.6.5	
Operating pressure	3.3.2	4.3.3.2	
Rated flow and pressure drop	3.3.5	4.3.3.4	
Proof pressure	3.3.6	4.3.3.5	
Fungus	3.6.3	4.3.6.3	
Operating pressure	3.3.2	4.3.3.2	
Rated flow and pressure drop	3.3.5	4.3.3.4	
Proof pressure	3.3.6	4.3.3.5	
Shock	3.6.6	4.3.6.6	4
Operating pressure	3.3.2	4.3.3.2	
Rated flow and pressure drop	3.3.5	4.3.3.4	
Proof pressure	3.3.6	4.3.3.5	

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TABLE I. Verification methods - Continued.

Title	Requirements	Verification	Sample No.
Vibration	3.6.7	4.3.6.7	
Operating pressure	3.3.2	4.3.3.2	
Rated flow and pressure drop	3.3.5	4.3.3.4	
Proof pressure	3.3.6	4.3.3.5	
Humidity	3.6.8	4.3.6.8	
Rated flow and pressure drop	3.3.5	4.3.3.4	4
Proof pressure	3.3.6	4.3.3.5	
Salt spray	3.6.4	4.3.6.4	
Operating pressure	3.3.2	4.3.3.2	
Rated flow and pressure drop	3.3.5	4.3.3.4	
Proof pressure	3.3.6	4.3.3.5	

1/ Each even numbered sample shall receive one environmental test, sequentially in the order listed in 4.3.6.

4.3.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 77 ± 15°F (30 ± 8°C).
b. Barometric pressure: 28.5 + 2 inches of mercury (Hg) (96 + 7 kPa).
c. Relative humidity: 50 ± 30 percent (%).

4.3.3 Operating requirements verification. Complete each verification under paragraph 4.3.3. Unless otherwise specified (see 6.2), performance testing shall be conducted under the following conditions:

a. Fluids. All tests shall be performed with hydraulic fluid (see 6.3). The hydraulic fluid temperature shall be 90 ±20°F (32 ± 11°C).
b. Cleaning. All grease or foreign material shall be removed from interior parts of the valve before testing.
c. Air removal. Air shall be completely removed from all hydraulic portions of the valves prior to tests.

4.3.3.1 Operating time. To determine conformance to 3.3.1, the valve shall be mounted and electrically connected as in intended use, with 18 Vdc applied. The valve shall be operated (change port interconnection), and the time required recorded.

4.3.3.2 Operating pressure. To determine conformance to 3.3.2, the valve shall pass the tests of 4.3.3.2.1 through 4.3.3.2.3, as applicable.

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4.3.3.2.1 Pressure port.

4.3.3.2.1.1 Type I. The valve shall be subjected to hydraulic pressure at the pressure port, with cylinder number 1 and 2 ports connected independently to loading valve and flow meter, and the return port open. With 250 psi (1724 kPa) maintained at the pressure port, the valve shall actuate with 18 Vdc or less applied to the solenoid. The pressure drop through the valve, either energized or de-energized, shall be not more than 150 psi (1034 kPa) at 6 gal/min (0.38 L/s) flow. With 3000 psi (20 685 kPa) maintained at the pressure port, the valve shall actuate with application of 18 Vdc or less, and the pressure drop through the valve, either energized or de-energized, shall be not more than 150 psi (1034 kPa) at 6 gal/min (0.38 L/s) (see 3.3.2, 3.3.3 and 3.3.5.1).

4.3.3.2.1.2 Type II. The valve in the energized position shall be subjected to a hydraulic pressure of 3000 psi (20 685 kPa) at port “A”, and port “B” connected to a flow meter (see 3.3.2). The valve shall then be energized, with an application of 18 Vdc, or less, actuating the solenoid (see 3.3.3). With the valve de-energized and a flow through the valve of 3.5 gal/min (0.22 L/s) , the pressure drop across the valve shall be recorded to determine conformance to 3.3.5. Test shall be repeated with set-up at port: “A” and “B” reversed.

4.3.3.2.2 Return port (type I only). The valve shall be energized with 18 Vdc, or less, at a pressure differential of 250 psi (1724 kPa) between the pressure port and the return port (pressure port at the greater procedure). A pressure of 1000 psi (6894 kPa) and a flow of 6 gal/min (0.38 L/s) shall then be applied at the return port. The pressure drop between the return port and cylinder number 1 port shall be not more than 150 psi (1034 kPa), with a flow of 6 gal/min (0.38 L/s) in either direction. The test shall be repeated, except that 1000 psi (6894 kPa) shall be applied at cylinder number 1 port (see 3.3.2, 3.3.3 and 3.3.5).

4.3.3.2.3 Actuation. With a hydraulic pressure of 3000 psi (20 680 kPa) at the pressure port and the valve operating as in 4.3.3.2.1, the valve shall be actuated with 24 Vdc no less than 10 times to assure that there is no indication of binding, chatter, or other unsatisfactory operation.

4.3.3.3 Operating voltage and current. During testing, to determine conformance to 3.3.3, voltage shall be varied between 18 and 30 Vdc and operation observed. To determine conformance to 3.3.4, current draw shall be measured during operation at 24 Vdc.

4.3.3.4 Rated flow and pressure drop. To determine conformance to 3.3.5, the rated flow shall be accurately maintained at the value specified in 3.3.5. A manometer or other accurate means of measuring pressure drop through interconnected ports shall be employed. The pressure drop shall be measured and recorded.

4.3.3.5 Proof pressure. To determine conformance to 3.3.6, the valve shall be subjected to the pressure specified in 3.3.6 at the pressure port and the return port (type I only) for a period

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of 2 minutes (min) at each application. The test shall be performed with the solenoid in both the energized and de-energized position. For type I valves, when testing the pressure port, the return port shall be open. The valve shall be electrically actuated after application of pressure. There shall be no evidence of external leakage, failure, or permanent deformation.

4.3.3.6 Burst pressure. After completion of all other tests, the valve shall be subjected to the pressure specified in 3.3.7 at the pressure and return port (type I only) for a period of 3 min at each application. The test shall be performed with the solenoid in both the energized and de-energized positions. For type I valves, when testing the pressure port, the return port shall be open. The rate of application of pressure shall not be more than 26000 psi (179 264 kPa) per minute. The valve need not be operable after testing. Results shall be recorded to determine conformance to 3.3.7.

4.3.3.7 Leakage.

4.3.3.7.1 Pressure port.

4.3.3.7.1.1 Type I. To determine conformance to 3.3.8.1.1, with the valve de-energized, a hydraulic pressure of 5 to 10 psi (35 to 69 kPa) shall be applied at the pressure port, and maintained for a period of 2 min. Leakage measurements (external and internal) shall be taken and shall not be more than the valves specified in 3.3.8.1.1. Pressure shall then be raised to 3000 psi (20 685 kPa), and maintained for a period of 2 min. Leakage measurements shall be recorded.

4.3.3.7.1.2 Type II. To determine conformance to 3.3.8.1.2, with the valve de-energized, the valve shall be subjected to a hydraulic pressure of 3000 psi (20 685 kPa) at port “A” and energized with an application of the not more than 18 Vdc, actuating the solenoid. A hydraulic pressure of 3000 psi (20 685 kPa) at port “A” shall be maintained for a period of 2 min, while the leakage at port “B” is recorded. The hydraulic pressure at port “A” shall then be reduced to 5 to 10 psi (35 to 69 kPa) and maintained for a period of 2 min, while the leakage at port “B” is recorded. The pressure set-up at ports “A” and “B” shall be reversed and test repeated.

4.3.3.7.2 Return port (type I only). With the valve de-energized and with the pressure port and cylinder number 1 port plugged, a hydraulic pressure of 5 to 10 psi (35 to 69 kPa) shall be applied at the return port and cylinder number 2 port. After test, the valve shall be examined to determine conformance to 3.3.8.2.

4.3.3.8 Solenoid temperature rise. To determine conformance to 3.3.9, the valve solenoid shall be subjected to 30 Vdc for a continuous 3 min, at an ambient temperature of 77 $\pm$ 15°F (30 $\pm$ 8°C).

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4.3.3.9 Dielectric strength. To determine conformance to 3.3.10, the solenoid shall be subjected to a voltage test of 1000 V rms at a frequency of 60 Hz between terminals and ground. Application shall be as specified herein, for 1 s for production units, and for 60 s for preproduction test units. The test voltage shall be raised from zero to the specified value as uniformly as possible, at a rate of approximately 500 volts rms per second, unless otherwise specified, the test voltage may be applied instantaneously during conformance testing. A voltmeter shall be used to measure the applied voltage to an accuracy of at least 5%. Suitable means shall be provided to indicate the occurrence of disruptive discharge, or an appropriate indicator light or an overload protective device may be used for this purpose. Flashover during this test, or a noticeable decrease in insulation resistance as a result of this test, shall constitute failure. Upon completion of the test, the test voltage shall be gradually reduced to avoid voltage surges. The dielectric withstanding voltage test should be used with caution particularly in conformance testing, as even an overpotential less than the breakdown voltage may injure the insulation and thereby reduce its safety factor. Therefore, repeated application of the test voltage on the same value is not recommended. Care should be taken to be certain that the test voltage is free of recurring transients or high peaks.

4.3.4 Interface requirements verification.

4.3.4.1 Overall envelope & interfaces inspection. Demonstrate placement of a valve in the present existing location, the valve shall interface electrically and mechanically without imperfections.

4.3.4.2 Weight. Weigh the valve to be 2 lb (0.9 kg) or less.

4.3.4.3 Solenoid. Inspect the solenoid for conformance as specified.

4.3.4.4 Coil. The coil shall be examined, and shall conform as specified.

4.3.4.5 Electromagnetic compatibility. To determine conformance to 3.4.5, the valve shall be tested as specified in MIL-STD-462 or equivalent (see 4.3.1).

4.3.5 Ownership and support requirements verification. Complete each verification under paragraph 4.3.5.

4.3.5.1 Endurance. To determine conformance to 3.5.1, the valve shall be solenoid operated for 20 000 cycles, 50% of which shall be with the solenoid voltage at 30 Vdc, with a hydraulic pressure of not less than 3000 psi (20 685 kPa) at the pressure port. A restriction shall be placed in the line from each cylinder port to maintain not less than a 3000 psi (20 685 kPa) at the cylinder port interconnected to the pressure port. For type I valve, the pressure at the opposite cylinder port shall drop to zero psi before proceeding to the next cycle. After test, the valve shall pass the tests specified in 4.3.3.2, 4.3.3.4, 4.3.3.5, and 4.3.3.7.

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4.3.5.2 Safety inspection. Inspect valve for hazardous burs, nicks, sharp edges, foreign materials, or other imperfections that pose physical danger to an installer/operator. Inspect valve for exposed, frayed, unsecured, or otherwise improperly protected electrical lines and connections that pose electrical danger to an installer/operator.

4.3.5.3 Interchangeability inspection. To determine conformance to 3.5.3, the interchangeability between different parts, subassemblies, and assemblies having the same military part number shall be verified.

4.3.5.4 Identification evaluation. Verify the presence of the required markings on the valve. After performing all environmental tests in 4.3.6, reinspect the valve markings for readability.

4.3.6 Operating environment requirements verification. Complete each verification under paragraph 4.3.6.

4.3.6.1 High temperature. To determine conformance to 3.6.1, the valve shall be subjected to the high temperature test specified in procedure II of MIL-STD-810, method 501.2, or equivalent (see 4.3.1). High temperature testing shall be conducted at the temperature specified in 3.6.1. While at the high temperature, the valve shall be subjected to tests specified in 4.3.3.2, 4.3.3.4, and 4.3.3.5. After the high temperature testing the valve shall be stabilized at room temperature, examined, and tested as specified in 4.3.5.1 (50 cycles only). Test results shall be recorded.

4.3.6.2 Low temperature. To determine conformance to 3.6.2, the valve shall be subjected to the low temperature tests specified in procedure II of MIL-STD-810, method 502.2, or equivalent (see 4.3.1), except the valve shall be maintained at a low temperature of not more than -65°F (-54°C). While at the low temperature the valve shall be tested as specified in 4.3.3.2, 4.3.3.4, and applicable tests of 4.3.3.5. After low temperature exposure, the valve shall be stabilized at room temperature, examined, and tested as specified in 4.3.5.1 (50 cycles only). Test results shall be recorded.

4.3.6.3 Fungus. To determine conformance to 3.6.3, the valve shall be subjected to the fungus resistance test of MIL-STD-810, method 508.3, or equivalent (see 4.3.1). After the test, the valve shall pass the tests as specified in 4.3.3.2, 4.3.3.4, and 4.3.3.5.

4.3.6.4 Salt spray. To determine conformance to 3.6.4, the valve shall be subjected to the salt spray test of ASTM B117, except the test duration shall be 200 hours. After the test, the valve shall pass the tests as specified in 4.3.3.2, 4.3.3.4, and 4.3.3.5.

4.3.6.5 Sand and dust. To determine conformance to 3.6.5, the valve shall be subjected to sand and dust resistance test of MIL-STD-810, method 510.2, or equivalent (see 4.3.1),

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except the test shall be conducted at an ambient temperature of 120 ± 5°F (49 ± 3°C) only, and the test period shall be 24 hr. The valve shall be mounted in the dust chamber as in intended use. After the test, the valve shall pass the tests specified in 4.3.3.2, 4.3.3.4, and 4.3.3.5.

4.3.6.6 Shock. To determine conformance to 3.6.6, the valve shall be subjected to the shock test specified in MIL-STD-810, method 514.1, or equivalent (see 4.3.1). After the test, the valve shall be examined and shall pass the tests specified in 4.3.3.2, 4.3.3.4, and 4.3.3.5.

4.3.6.7 Vibration. To determine conformance to 3.6.7, the valve shall be mounted as specified using suitable mounting apparatus to assure that mounting is free from resonances over the test frequency range. The valves shall be subjected to a simple harmonic motion having an amplitude of 0.03 in. (0.06 in. maximum total excursion), the frequency being varied uniformly between the approximate limits of 10 and 55 Hz. The entire frequency range, from 10 to 55 Hz and return to 10 Hz, shall be traversed in approximately 1 min, the vibration shall be applied for 1 hr at 77 ± 15°F (25 ± 8°C) and 30 min at -65 ± 5°F (-54 ± 3°C), in the direction of each of the three major axes (total of 4-1/2 hr). This test shall be made under electrical-load conditions. Following the vibration test, the valve shall be examined and shall pass the tests specified in 4.3.3.2, 4.3.3.4, and 4.3.3.5.

4.3.6.8 Humidity. To determine conformance to 3.6.8, the valve shall be subjected to the humidity test specified in MIL-STD-810, method 507.2, procedure III, or equivalent (see 4.3.1). After the test, the valve shall pass the tests specified in 4.3.3.4 and 4.3.3.5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. These valves are intended for use with loader-transporter, guided missile (Hawk Loader). Valves are military unique because they must operate under pressure and environmental conditions which exceed any commercial valve characteristics.

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6.1.1 Type I. The valve covered by this specification is a two position, four-way valve, used to automatically select the fluid path in a 3000 psi (20 685 kPa) hydraulic system to reduce the final cylinder actuating pressure.

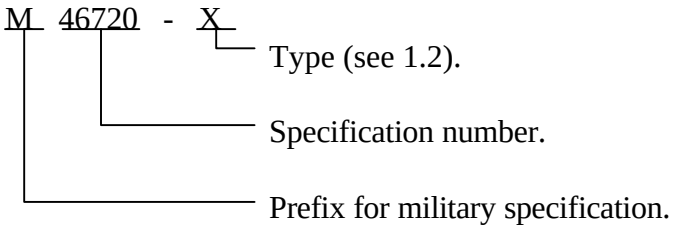
6.1.2 Type II. The valve covered by this specification is a two position, two-way valve, used to interrupt the fluid flow in a 3000 psi (20 685 kPa) hydraulic system, or permit flow in either direction through the valve.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title. number, and date of this specification.
b. Type of valve required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article samples should be furnished (see 3.1).
e. Serial numbers, if required (see 3.5.4).
f. If preproduction inspection is required (see 4.1.1.1).
g. If initial production inspection is not required (see 4.1.1.2).
h. Test conditions if other than as specified (see 4.3.2).
i. If performance conditions are other than specified (see 4.3.3).
j. Packaging requirements (see 5.1).

6.3 Recommended service products. Typically, the valves should be serviced with MIL-H-6083 or MIL-H-5606 as a hydraulic fluid.

6.4 Part or Identifying Number (PIN). The PIN to be used for valves acquired to this specification are created as follows:



6.5 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously

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acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.6 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skills and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table I and their designated frequency based on a risk assessment for the procurement.

6.7 Disposition of test assemblies. Assemblies undergoing destructive tests should not be used and should be indelibly marked "DO NOT USE."

6.8 Subject term (key word) listing.

Burst pressure
Electrically activated
Flow capacity
Hydraulic directional control
Leakage
Operating pressure
Return port

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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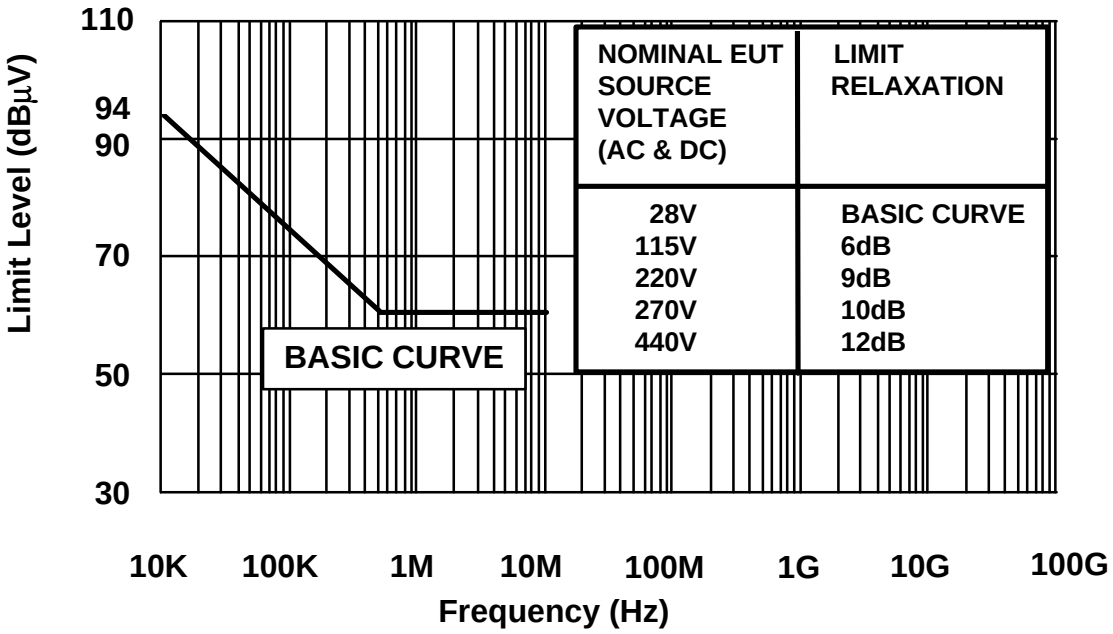


FIGURE 1. Conducted emissions levels.

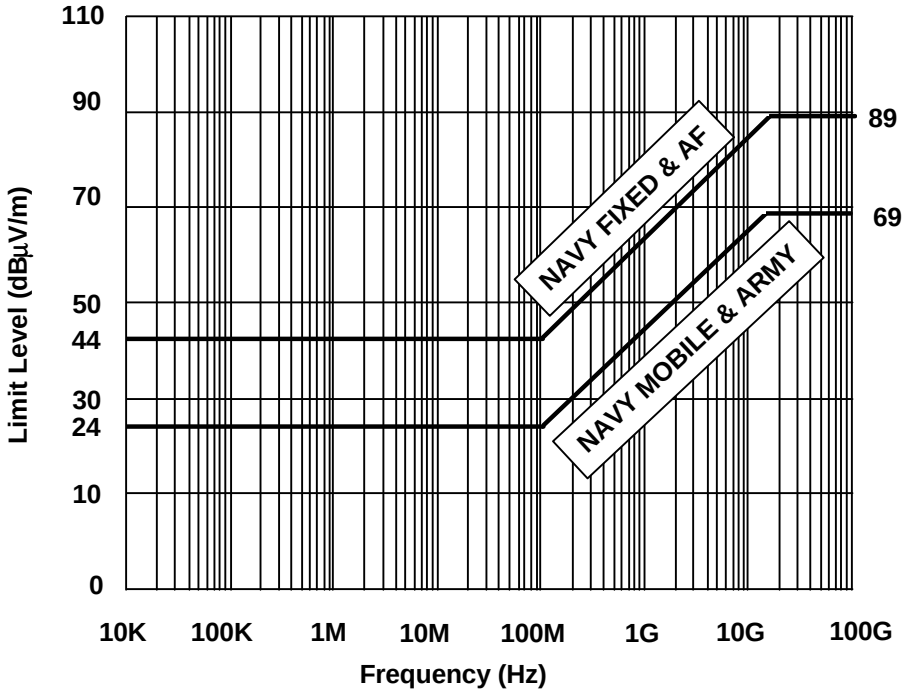


FIGURE 2. Radiated emissions levels.

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Preparing Activity:
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Review Activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-46720C	**2. DOCUMENT DATE *(YYMMDD)*** 971215

3. DOCUMENT TITLE
Valve, Solenoid

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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